

Skills Summary

Reading and Plotting Points on a Grid

Use this reference while completing your worksheet or when studying.

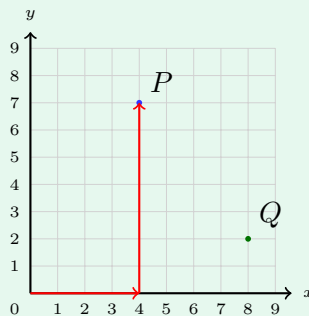
Skill 1 — Reading a point: across first, then up

An ordered pair is written (x, y) . The first number says how far to travel **across** from the corner marked 0; the second says how far to travel **up**.

Always in that order. $(3, 7)$ and $(7, 3)$ use the same two numbers and land in completely different places, which is why they are called *ordered* pairs.

A coordinate of 0 is a real instruction: it means *do not travel that way at all*. The corner itself is $(0, 0)$.

Example: Write point P as an ordered pair.



Step 1. A point is read as a journey from the corner, and the journey always goes across before it goes up — so trace the red arrows in that order.

Step 2. Across the bottom to sit under P : four squares. Then straight up to P : seven squares.

$$\Rightarrow P = (4, 7)$$

Try it: Write the green point Q on the same grid as an ordered pair.

$(8, 2) = Q$: check

Skill 2 — Plotting a point and measuring a line

To plot (x, y) : put your finger on 0, slide across x squares, then go up y squares, and mark the spot.

To measure a straight line between two points: if the line is *upright*, both points have the same x , so subtract the two y -coordinates. If the line is *flat*, both points have the same y , so subtract the two x -coordinates. Only one of the two coordinates does any work.

Example: Plot $R(5, 2)$ and $S(5, 7)$. How many squares long is $\overline{RS}$?

Step 1. Both points are five squares across, so the line joining them is upright and only the up-and-down numbers can differ — subtract those.

Step 2. y of S minus y of R : $7 - 2$.

$\Rightarrow \overline{RS} = 5$ squares

Try it: Plot $T(3, 3)$ and $U(3, 9)$. How many squares long is $\overline{TU}$?

check: 9 squares

Skill 3 — Two rules make a list of points

Run two counting rules side by side. The number from the first rule is the x , the number from the second rule is the y , and together they make one ordered pair.

Plot the pairs in order and they fall on a straight line. If one rule adds 1 each time and the other adds 3, then y climbs three times as fast as x — so y is always three times x , and you can jump straight to any pair without writing out the rows in between.

Example: Rule A: start at 0, add 1. Rule B: start at 0, add 3. What is y when $x = 2$?

Step 1. Rule B climbs three steps for every one step of Rule A, so instead of listing rows, multiply the x by three.

Step 2. The pairs run $(0, 0)$, $(1, 3)$, $(2, 6)$, and $3 \times 2 = 6$ agrees.

$\Rightarrow y = 6$, so the pair is $(2, 6)$

Try it: Same two rules. What is y when $x = 4$?

check: $12 = 4 \times 3$, so the pair is $(4, 12)$

(!) Watch out: the commonest mistake on a grid is doing the two journeys in the wrong order — up first, then across. Say the pair out loud as you plot it: “across four, up seven.” If a point lands somewhere unexpected, check the order before checking anything else.